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Subject: Introduction to Web Development

Semester: 5th

Assignment 1

Q1. Difference between frontend, backend and full-stack development with examples

Frontend development means the part of a website that we can see and use directly, like the buttons, images, menus and overall design. It is made using HTML, CSS and JavaScript. Backend development is the part that we don't see, it works behind the scenes and handles things like storing data, login system and processing requests. It is usually made using languages like Python, PHP, Node.js etc along with a database.

Full stack development means the developer knows both frontend and backend and can build the complete website on his own.

Example - In Amazon website, the part where we see products, pictures and add to cart button is frontend. The part which checks if the product is in stock and processes our payment is backend. A person who can do both is a full stack developer.

Q2. Diagram of client-server model

```
   CLIENT                      SERVER
  (Browser)                 (Web Server)
      |                          |
      |----- request page ----->|
      |                          |
      |<---- sends back page ---|
      |                          |
```

Here the client (our browser) asks the server for a webpage, and the server sends the page back which we can then see in our browser.

Q3. How browser requests and displays a webpage

First we type the website address in the browser. The browser then finds the IP address of that website using DNS. After that it connects to the server and asks for the webpage using an HTTP request. The server checks the request and sends back the html, css and js files needed. Finally the browser reads all these files and shows the final webpage to us on the screen.

Q4. Tools required for web development environment

Some common tools that are needed:

- A code editor like VS Code, to write the code
- A web browser like Chrome or Firefox, to check the output
- Git, used to keep track of changes made in the code
- GitHub, used to store the code online and share it with others
- Node.js, needed to run JavaScript outside of the browser
- A local server / live server extension, to see how the website looks before uploading it online

Q5. What is a web server, examples

A web server is basically a program that stores webpages and sends them to users whenever they ask for it through their browser. It works using HTTP protocol.

Some common web servers are Apache, Nginx, and Microsoft IIS.

Q6. Roles of frontend developer, backend developer, database administrator

Frontend developer - designs and builds the part of the website that users can see and interact with.

Backend developer - handles the server side, writes the logic and connects the website with the database.

Database administrator - takes care of the database, makes sure the data is safe, organized and can be accessed properly.

Q7. Install VS Code and configure for HTML, CSS, JS (screenshot)

I downloaded VS Code from the official website code.visualstudio.com and installed it on my laptop. After installing, I opened the extensions tab and added the Live Server extension, which helps to see the webpage output directly without uploading it anywhere. Then I created a new folder with an index.html file inside it and opened that folder in VS Code to start writing code.

(Screenshot of my VS Code setup is attached below)

Q8. Static vs dynamic websites with example

Static websites are those where the content stays the same for everyone and doesn't change unless someone edits the code manually. They are made using plain HTML and CSS. Example - a simple portfolio website.

Dynamic websites are those where the content can change depending on the user or the situation, like showing different products or updating in real time. These use a database and backend language. Example - a shopping website like Flipkart, where the products and prices keep changing.

Q9. Five web browsers and their rendering engines

Google Chrome - uses Blink engine.

Mozilla Firefox - uses Gecko engine.

Safari - uses WebKit engine.

Microsoft Edge - earlier had its own engine, now uses Blink (same as chrome).

Opera - also uses Blink engine.

Even though some of these browsers use the same rendering engine, they can still show slight differences because of extra features added by each browser and how they handle things like extensions or settings.

Q10. Diagram of web architecture flow (client, server, database, APIs)

```
                      EXTERNAL API
                           |
                           |
   CLIENT  <---------->  SERVER  <---------->  DATABASE
  (Browser)             (Logic)                (Stores data)
```

The client sends a request to the server. The server has all the logic of the website, so it processes the request and if needed, gets data from the database. Sometimes the server also needs to contact an external API (like a payment gateway or maps service) to complete the request. After everything is done, the server sends the final response back to the client.